

Covert operators are picked to minimize QuD-Ambiguity: The view from *pex* and *only*¹

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Abstract. Silently derived scalar implicatures (SI) have a semantic effect similar to that of the overt operator *only*, differing primarily in how the relevant meaning components are distributed between presupposition and assertion (Fox 2007; Bassi et al. 2021 i.a.). Why is the division of labor between silent SIs and overt *only* so stable across languages? This paper argues that the answer lies in ambiguity minimization. Specifically, it argues that (i) SIs and *only* are used to address distinct Questions under Discussion (QuD; Roberts 2012 i.a.); (ii) because covert operations are optional, they create uncertainty about which QuD should be accommodated; (iii) when two operations are semantically similar enough, they compete for covertness, such that the covert operation (if any) should be the one that minimizes a QuD-related measure of ambiguity. This predicts SIs to be the preferred covert operation. The proposal provides a principled explanation for the semantic asymmetry between SIs and *only* and, in turn, yields an indirect argument for a grammatical, presuppositional view of SIs.

Keywords: scalar implicatures, exhaustification, Question under Discussion, covertness, parsing ambiguities.

1. Introduction: Two approaches to scalar implicatures, and a common issue

Scalar implicatures (henceforth **SI**s) arise when a speaker chooses a weaker term (e.g. *some*, see (1a)) on a scale of informational strength, implying that a stronger alternative (e.g. *all*, see (1b)) does not hold; see (1c). A distinctive feature of SIs is that they are often optional and can be easily canceled (Grice, 1989); see (1d).

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|-----|--|-----------------------------|
| (1) | a. Jo ate some of the cookies. | <i>Uttered Prejacent</i> |
| | b. Jo ate all of the cookies. | <i>Stronger Alternative</i> |
| | c. (1a) $\rightsquigarrow$ Jo did not eat all of the cookies. $\equiv \neg$ (1b) | <i>Scalar Implicature</i> |
| | d. Jo ate some of the cookies; in fact he ate all . | <i>Canceled SI</i> |

Within the Neo-Gricean framework (Horn, 1972; Grice, 1975; Gazdar, 1979; Horn, 1984; Sauerland, 2004, i.a.), SIs are taken to be drawn by the pragmatic module of the grammar (i.e. post-compositionally), based on core principles governing cooperative conversations: truthfulness, informativeness, opinionatedness.² By contrast, within the grammatical framework (Chierchia et al., 2009, 2011; Magri, 2011; Crnič, 2012; Meyer, 2013, 2016; Bar-Lev and Fox, 2017; Fox, 2018; Bassi et al., 2021, i.a.), SIs result from the insertion of a covert syntactic operator often dubbed *exh* (for “exhaustification”). Semantically, *exh* can be taken to conjoin its prejacent (*p*)

¹I would like to thank Amir Anvari, Athulya Aravind, Josh Boivin and Nina Haslinger, as well as the audience of SuB30 for relevant questions and feedback.

²In Sauerland (2004), (un)opinionatedness helps derive the optionality of SIs.

with the negation of non-weaker alternatives belonging to a set $\mathcal{A}_p$; see (2).³ The meaning of *exh* is thus close to that of *only*, as illustrated in (3).

- (2) $\llbracket exh \rrbracket = \lambda p. p \wedge \forall q \in \mathcal{A}_p : p \not\leq q. \neg q$
 (3) $\llbracket exh(\text{Jo ate some of the cookies}) \rrbracket$
 $\equiv \llbracket \text{Jo ate some of the cookies} \rrbracket \wedge \neg \llbracket \text{Jo ate all of the cookies} \rrbracket$
 $\cong \llbracket \text{Jo ate only some of the cookies} \rrbracket$

The grammatical framework can capture the optionality of SIs by positing that *exh* is optionally merged.⁴ Therefore, a sentence in which *exh* can be inserted in one site is in principle ambiguous between two parses: a parse with *exh*, and a parse without. Though the grammatical view proved successful in many subdomains of pragmatics, the existence of *exh* raises an obvious question: does *exh* semantically differ from *only*, and if so, in what sense? There are several empirical reasons to assume that *exh* and *only* convey the same overall information, yet cut the pie differently between presupposition and assertion. Under Bassi et al.’s (2021) view, *exh* asserts its prejacent and presupposes the negation of non-weaker alternatives. *Exh* was hence dubbed *presuppositional*, and abbreviated *pex*; see (4). The remainder of this paper will adopt this view.⁵ *pex*’s presuppositions (i.e. SIs) are taken to be easily *accommodated*, i.e. added to the Common Ground, should they not be met in context (Stalnaker 1974 i.a.). By contrast, *only* is often taken to presuppose its prejacent, and to assert the negation of non-weaker alternatives; see (5).⁶

- (4) $\llbracket pex \rrbracket = \lambda p. \underline{\forall q \in \mathcal{A}_p : p \not\leq q. \neg q}. p$ (5) $\llbracket only \rrbracket = \lambda p. \underline{p}. \forall q \in \mathcal{A}_p : p \not\leq q. \neg q$

Independently of the specific differences between *pex* and *only*, a second question arises: why is *only* robustly overt cross-linguistically, while *exh/pex* is not?⁷ Specifically, why do overt exhaustifiers robustly presuppose their prejacent, while covert ones—if they are real—do not? This puzzle in fact extends to Neo-Gricean approaches for which there is no such thing as a covert exhaustifier—just a covert pragmatic process: why does this process never background the proposition under consideration, while appealing to *only* does? Put differently, it is not because the Neo-Gricean view posits that SIs and *only* operate in different realms, that it can explain *why* these two operations divide asserted and backgrounded meaning the way they do.

This paper aims to explain the missing link between (c)overttness and the presupposition-assertion divide in the case of SIs and *only*. It argues that the (c)overttness asymmetry between

³More sophisticated semantic entries for *exh* restrict the set of negated alternatives to *Innocently Excludable* ones (Fox, 2007), and may also assert *Innocently Includable* alternatives (Bar-Lev and Fox, 2017). Here we focus on cases in which these more sophisticated models are not directly needed.

⁴Alternatively, one can assume that this operator is always merged, but that the alternatives it operates on can be impoverished by *pruning* (Fox and Katzir, 2011; Ivlieva, 2012; Katzir, 2014; Denić, 2023, i.a.).

⁵*exh* also differs from *only* in that it makes *both* its prejacent and additional SIs at-issue. As we will see, however, this characterization appears less empirically adequate once QuDs are taken into account, and would moreover make the present proposal less straightforward to pursue.

⁶Though see Atlas (1993); Horn (2002); Roberets (2011); Crnić (2024) for more elaborate views on *only*.

⁷One could argue that *pex* may come with overt features, such as pitch accent, and that clefts may in certain respects instantiate *overt* versions of *pex*. This will not be fully ruled out by our approach, which only predicts that, if *pex* or *only* is covert, then it has to be *pex*.

SIs and *only*⁸ can be most easily explained assuming Bassi et al.’s view. On this view, the asymmetry follows from a more general pressure to minimize ambiguity. Ambiguity is understood here relative to the (implicit) Question under Discussion (henceforth **QuD**; Buring 2003; Roberts 2012; Ginzburg 2012; Riester 2019; Zhang 2022 i.a.). The key new claim is that, when two operators convey the same overall information, the choice of which one, if any, should remain covert is driven by the minimization of QuD-ambiguity.

This paper is structured as follows. Building on the view that overt and covert exhaustifiers differ in how they divide assertion and presupposition, Section 2 argues that this difference leads speakers to infer different implicit QuDs from structures containing such operators. Section 3 then shows that covert operators, if optional, induce an ambiguity regarding which implicit QuD should be inferred. A measure of QuD-ambiguity is then introduced, according to which *pex* induces less QuD-ambiguity than a putative covert *only* (dubbed *conly*), thereby favoring *pex* as the optimal covert operator. Section 4 extends the proposal to more complex QuD configurations. Section 5 discusses broader implications of the proposal and concludes the paper.

2. Exhaustifiers and the (implicit) QuD

2.1. Felicitously addressing QuDs

The Context Set, henceforth **CS** (Stalnaker, 1978) standardly refers to the set of worlds compatible with the premises of the conversation at stake. QuDs are identified with partitions of the CS (Hamblin, 1958; Groenendijk and Stokhof, 1984), i.e. sets of non-empty, disjoint subsets of the CS (*cells*), which together cover it. Given any set of propositions (called *alternatives*), a partition of the CS can be derived by grouping into cells the worlds of the CS that agree on all the propositions at stake. We will refer to such partitions as Hamblin-style partitions.

We assume that accommodating a presupposition *p* on a (partitioned) CS amounts to intersecting it with *p* (Stalnaker, 1970, 1974; Karttunen, 1974; Heim, 1983). We also build on the crucial observation that accommodated presuppositions should not settle the QuD (Heim, 2015; Aravind et al., 2022; Doron and Wehbe, 2024). Doron and Wehbe’s implementation of this principle, given in (6), states that an utterance’s assertion should be able to rule out a cell of the partitioned CS obtained *after* accommodating the utterance’s presupposition.

- (6) **POST ACCOMMODATION INFORMATIVITY** (Doron and Wehbe, 2024). A sentence *S* presupposing *p* can be felicitously asserted in a CS *C* partitioned by a set of alternatives *Q* only if *S*’s assertion is disjoint from at least one cell of the partition induced by *Q* on $C \cap p$ (CS obtained after accommodating *p*).

A useful consequence of this principle is that accommodation should not produce a one-cell QuD, since such a configuration prevents any non-contradictory assertion from eliminating a cell (see Figure 1a and 1b). By contrast, Figures 1c and 1d illustrate accommodation configurations that are not ruled out by Doron and Wehbe’s constraint. The next section examines how this constraint applies to scalar items such as *some* and *all*, which may carry *pex*-induced presuppositions.

⁸More precisely: between the *meanings* typically conveyed by SIs and by *only*.

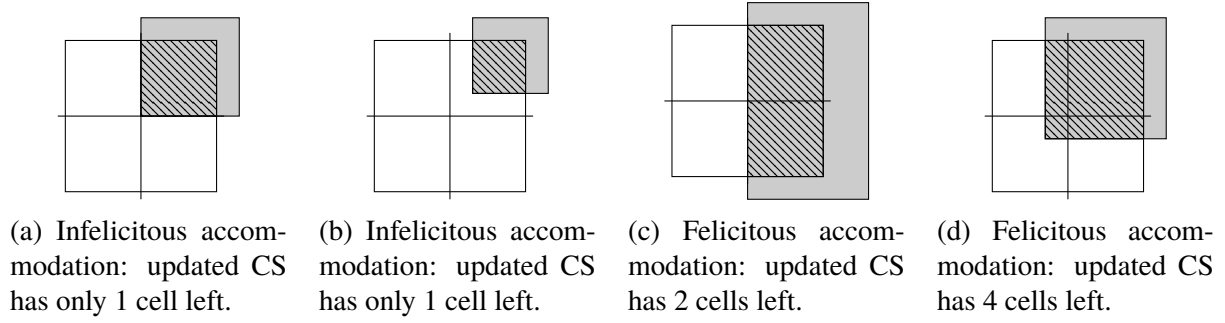


Figure 1: (In)feliciously accommodating p on a 4-cell partition of the CS. Grey rectangles define p . The hatched regions define the partitioned CS after accommodating p .

2.2. Only some and *exh* some address distinct QuDs

Assuming the existence of *pex*, and the entries for *pex* and *only* in (4) and (5) respectively, the question-answer pairs in (7) and (8) comply with the principles we just introduced.⁹

- (7) Did Jo eat **all** of the cookies or not? (8) Did Jo eat **some** of the cookies or not?
- QuD: $\forall \quad \neg\forall$ QuD: $\exists \quad \neg\exists$
- Jo ate { all[✓], some[✗], only some[✓] } Jo ate { all[?], some[✓], only some[✗] }

Let us first examine (7). Assuming the QuD in (7) partitions the CS into $\forall$ and $\neg\forall$ worlds, answering with *all* is fine, because *all* is presuppositionless and rules out the $\neg\forall$ -cell of the QuD, thus relevantly addressing it (Figure 2a). Answering with *some* however, is infelicitous. Here is why. If *some* is understood literally as $\exists$ (presuppositionless, *pex*-less parse), the answer does not rule out any cell, and so does not relevantly address the QuD (Figure 2b). Understanding *some* as *pex*($\exists$) does not help: in that case, the $\neg\forall$ -cell is identified presuppositionally, which does not leave enough possibilities open for the assertion ($\exists$) to rule out a cell (Figure 2c).¹⁰ So either way, *some* does not constitute a felicitous answer. Lastly, answering with *only some* is fine, because the at-issue component of this answer ($\neg\forall$) rules out a cell, even after the presupposition ($\exists$) gets accommodated (Figure 2d).

The same kind of reasoning applies to the QuD in (8), which can be taken to partition the CS into $\exists$ and $\neg\exists$ worlds. Answering with *all* is slightly degraded, probably due to overinformativeness (Figure 3a). Answering with *some* is fine, because it rules out the $\neg\exists$ -cell, whether or not a $\neg\forall$ presupposition (contributed by *pex*) gets accommodated (Figures 3b and 3c). Lastly, answering with *only some* is degraded, because it identifies the $\exists$ -cell presuppositionally, leaving no space for the assertion ($\neg\forall$) to be relevant (Figure 3d).

⁹A similar point is made in Hénot-Mortier (2026) in the context of redundant “scalar” disjunctions.

¹⁰If *exh* had been assumed instead, answering with *some* could be understood as *asserting* $\exists \wedge \neg\forall$, which would be relevant though overinformative: $\exists \wedge \neg\forall$ strictly entails the $\neg\forall$ -cell. We would thus predict that answering *some* to a *whether all* QuD should be roughly as felicitous as answering *all* to a *whether some* QuD. But the former move is more degraded and “feels” different than the latter. This observation favors a *pex* analysis over an *exh* analysis.

Covert operators are picked to minimize QuD-Ambiguity: The view from *pex* and *only*

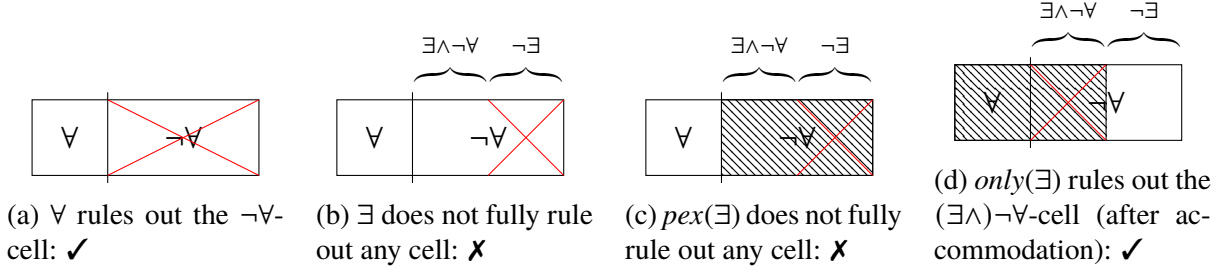


Figure 2: (In)feliciously answering a *whether all* QuD. Hatched regions define the partitioned CS obtained after accommodating the answer’s presupposition (if any). Crossed-out regions are those ruled out by the answer’s assertion (after accommodation, whenever applicable).

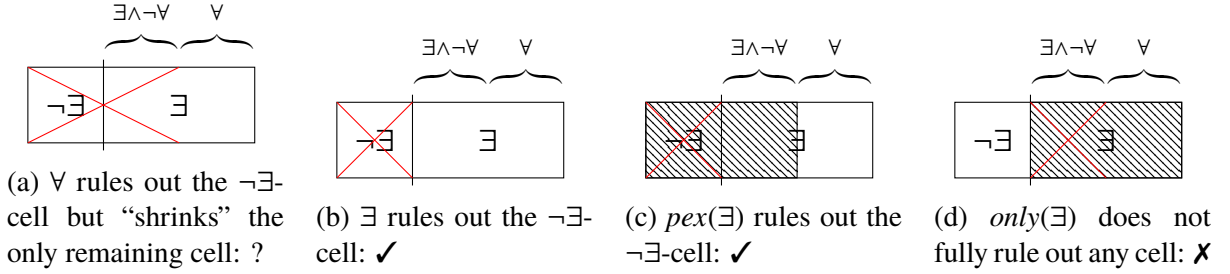


Figure 3: (In)feliciously answering a *whether some* QuD. Same conventions as Figure 2.

2.3. *Only some* and *exh some* evoke distinct QuDs

The previous section illustrated the compatibility between overt questions and assertions. Building on Roberts (2012); Riester (2019); Zhang (2022) i.a., we now argue that an assertion may *evoke* the implicit QuD(s) it could felicitously address. The patterns in (7-8) then suggest that utterances conveying *all* and *only some* may evoke a QuD of the form $\{\forall, \neg\forall\}$, while an utterance conveying *some* may evoke a QuD of the form $\{\exists, \neg\exists\}$. There are of course many other possibilities, depending on e.g. focus structure.¹¹ For present purposes, we restrict attention to two-cell “polar” QuDs.

Building on Hénót-Mortier (2025, 2026) we assume that evoked QuDs are defined on the CS *updated with whatever presupposition the utterance carries*. So, an assertion A presupposing p , will evoke a QuD whose cells are inferred from A , but defined on the CS intersected with p . A and $\neg A$ are taken to evoke the same implicit QuDs. This predicts that utterances of *all* (now abbreviated $\forall$), literal *some* (now $\exists$), *pex some* ($pex(\exists)$), and *only some* ($only(\exists)$) evoke distinct QuDs—see Table 1. Specifically, $\exists$ and $pex(\exists)$ evoke the same QuD “structure”, but $pex(\exists)$ defines it on a smaller CS entailing its presupposition ($\neg\forall$). Likewise, $\forall$ and $only(\exists)$ evoke the same QuD structure, but $only(\exists)$ defines it on a smaller CS entailing its presupposition ($\exists$).

¹¹For instance, focus on *cookies* in (1), may induce a partition of the CS according to what Jo ate. It can be shown however that considering these alternative QuDs does not fundamentally affect our subsequent predictions, because such QuDs may be equally raised by utterances involving *pex* and *only* at the same insertion site.

Table 1: QuDs inferred from utterances involving *some* and *all*, with or without *pex/only*.

	Parse	Assertion	Presupposition	QuD
(i)	$\forall$	$\forall$	-	$\boxed{\forall \quad \neg\forall}$
(ii)	$\exists$	$\exists$	-	$\boxed{\exists \quad \neg\exists}$
(iii)	$pex(\exists)$	$\exists$	$\neg\forall$	$\boxed{\exists \wedge \neg\forall \quad \neg\exists}$
(iv)	$only(\exists)$	$\neg\forall$	$\exists$	$\boxed{\forall \quad \neg\forall \wedge \exists}$

It thus appears that $\exists$ and $pex(\exists)$ share the same evoked QuD structure, while $\exists$ and $only(\exists)$ do not. Building on this contrast, the next section argues that covert operators give rise to ambiguities regarding which QuD should be inferred, and specifically shows that *pex* creates less QuD-ambiguity than a covert counterpart of *only*.

3. Parsing ambiguities trigger QuD ambiguities

3.1. Covert operators generate QuD ambiguities

A sentence in which *pex* can be inserted in one site is in principle ambiguous between two parses: a parse with *pex*, and a parse without. *pex*-able sentences in turn induce an ambiguity regarding which implicit QuD should be inferred. An utterance of *some* for instance, may evoke a QuD addressed by $\exists$ (typically, $\{\exists, \neg\exists\}$, see Table 1-(ii)) or by $pex(\exists)$ (typically $\{\exists, \exists \wedge \neg\forall\}$, see Table 1-(iii)). Of course, this kind of ambiguity does not arise with overt operators, like *only*. The possible QuDs evoked by *all*, *some*, and *only some* are summarized in Table 2.

 Table 2: Attested case: QuDs evoked by utterances containing *all*, *some* (ambiguous between $\exists$ and $pex(\exists)$), and *only some*.

Utterance	Parse	QuD
<i>all</i>	$\forall$	$\boxed{\forall \quad \neg\forall}$
<i>some</i>	$\exists$	$\boxed{\exists \quad \neg\exists}$
	$pex(\exists)$	$\boxed{\exists \wedge \neg\forall \quad \neg\exists}$
<i>only some</i>	$only(\exists)$	$\boxed{\forall \quad \neg\forall \wedge \exists}$

 Table 3: Unattested case: QuDs evoked by utterances containing *all*, *some* (ambiguous between $\exists$ and $conly(\exists)$), and *opex some*.

Utterance	Parse	QuD
<i>all</i>	$\forall$	$\boxed{\forall \quad \neg\forall}$
<i>some</i>	$\exists$	$\boxed{\exists \quad \neg\exists}$
	$conly(\exists)$	$\boxed{\forall \quad \neg\forall \wedge \exists}$
<i>opex some</i>	$opex(\exists)$	$\boxed{\exists \wedge \neg\forall \quad \neg\exists}$

What would happen if *pex* were overt, and *only*, covert? We define *opex* as the putative overt realization of *pex*, and *conly* as the putative covert realization of *only*. We assume that languages are endowed with at most one covert exhaustifier,¹² *pex* or *conly*. In a language with *opex* and *conly*, an utterance of *some* may be parsed as $\exists$ or $conly(\exists)$, the latter presupposing $\exists$, and asserting $\neg\forall$. This induces an ambiguity between a $\{\exists, \neg\exists\}$, and a $\{\forall, \exists \wedge \neg\forall\}$ QuD.

¹²Otherwise, it would be impossible to infer the status of SIs-at-issue or presuppositional.

Additionally, an utterance of *opex some* would not lead to parsing-induced QuD-ambiguity, since *opex* is posited overt. This is summarized in Table 3. A language with both overt operators *opex* and *only* on the other hand, would be characterized by minimal operator-induced QuD-ambiguity, but at the cost of increased prolixity. The next section defines a relative measure of QuD-ambiguity, and shows that the state of affairs described in Table 2, which is the cross-linguistically attested one, leads to comparatively less QuD-ambiguity than the unattested state of affairs described in Table 3.

3.2. Covert operators can be compared w.r.t their potential for QuD-ambiguity

When an utterance may contain a covert operator, e.g. *pex*, or *conly*, an ambiguity arises regarding the implicit QuD(s) inferred from the utterance. In the attested case (Table 2), the two possible QuDs produced by the parsing ambiguity share a cell (the $\neg\exists$ -cell), while in the unattested case (Table 3), they do not. In that sense, the attested configuration induces less ambiguity than the unattested configuration: in the former, at least we know that the QuD evoked by a covertly exhaustifiable utterance involves a $\neg\exists$ -cell.

More generally, if U and U' are two utterances, then U is less QuD-ambiguous than U' if the questions evoked by (the different parses of) U are “closer” to each other, than those evoked by U' are close to each other. Closeness is understood in terms of sets of matching cells: “close” questions have a lot of cells in common. A set of questions Q is then less ambiguous than another set of questions Q' , if the set of cells that all members of Q have in common, *contains* the set of cells that all members of Q' have in common. (9) implements this in the form of a partial order between sets of questions. The set of cells that questions have in common corresponds to their grand intersection; see (10).

(9) **RELATIVE QUD-AMBIGUITY BETWEEN SETS OF QUESTIONS.** A set of questions Q is less ambiguous than another set Q' ($Q \prec Q'$) iff $M(Q) \supset M(Q')$

(10) **MATCHING CELLS.** Let Q be a set of questions. $M(Q) = \{c \mid \forall Q \in Q. c \in Q\} = \bigcap_{Q \in Q} Q$

Based on this idea, a relative measure of QuD-ambiguity induced by covert operators such as *pex* or *conly* can be defined. (11) first defines the set of questions induced by the presence or absence of a given covert operator at a fixed insertion site, in a fixed utterance. Building on (9) and (11), (12) then introduces a relative measure of QuD-ambiguity between covert operators, inserted at a fixed site. This measure makes it possible to compare covert operators in terms of their potential to induce QuD-ambiguity.

(11) **QUESTIONS INDUCED BY A COVERT OPERATOR.** Let U be an utterance, Φ a covert operator and S a site in U where Φ could be inserted. Let U' be the result of the insertion of Φ at site S in U . The questions induced by the covertness of Φ at site S in U , noted $Q(U, S, \Phi)$ corresponds to the set of questions evoked by either U or U' .

(12) **COMPARING COVERT OPERATORS RE:QUD-AMBIGUITY.** Let U be an utterance, Φ and Ψ covert operators and S a site in U where Φ or Ψ could be inserted. Φ induces less QuD-ambiguity than Ψ at site S in U iff $Q(U, S, \Phi) \prec Q(U, S, \Psi)$.

Let us first review a simple example with $\Phi=pex$ and $\Psi=only$ in a simple matrix environment like the *some*-utterance in (1a). Considering *pex* first, applying (11) to this utterance yields the set of questions evoked by $\exists$ and $pex(\exists)$, i.e. $Q(some, root, pex)$ is the set $\{\{\exists, \neg\exists\}, \{\exists \wedge \neg\forall, \neg\exists\}\}$. The two questions in this set share the $\neg\exists$ -cell, meaning, $\mathcal{M}(Q(some, root, pex)) = \{\neg\exists\}$. Now, considering *only*, applying (11) to the same *some* utterance yields the set of questions evoked by $\exists$ and $only(\exists)$, i.e. $Q(some, root, only)$ is the set $\{\{\exists, \neg\exists\}, \{\forall, \neg\forall \wedge \exists\}\}$. The questions in this set do not have any cell in common, meaning, $\mathcal{M}(Q(some, root, only)) = \emptyset$. Because $\emptyset \subset \{\neg\exists\} = \mathcal{M}(Q(some, root, pex))$, we conclude $Q(some, root, pex) \prec Q(some, root, only)$ as per (9). *pex* therefore induces less QuD-ambiguity than *only* in the context of a simple *some*-utterance, as per (12). The next section attempts to generalize this claim to the level of an entire language—providing a metric according to which *pex* is overall less QuD-ambiguous than *only*.

3.3. Semantically similar operators “compete” for covertness

To compare the *general* potential for QuD-ambiguity of two covert operators, the result established in the previous section for matrix *some* (interpreted as $pex(\exists)$ or $only(\exists)$) must be extended to the language as a whole. Ultimately, this should explain why certain covert operators, such as *pex*, deserve to be part of the grammar of a language, while others, such as *only*, do not. The guiding intuition is that languages minimize overall QuD-ambiguity when selecting, among semantically similar covert operators, which one to grammaticize. Specifically, (13) proposes that a covert operator inducing comparatively less QuD-ambiguity (as measured by $\prec$) than a competitor with similar meaning should be preferred by the grammar. Since $\prec$ is a partial order, an optimal covert operator need not outperform its competitor in every case. Rather, it must be *sometimes* better (13i), and *never* worse (13ii), than its competitor, when it comes to QuD-ambiguity.

- (13) **MINIMIZE QUD-AMBIGUITY.** Let Φ and Ψ be two covert operators s.t. for any utterance U , inserting Φ or Ψ at the same site, leads to the same CS after presupposition and assertion have been processed. Then, Φ should be preferred by the grammar iff:
- (i). Φ sometimes induces less QuD-ambiguity than Ψ , meaning, there is an utterance U and a site S in U in which Φ or Ψ can be inserted, s.t. $Q(U, S, \Phi) \prec Q(U, S, \Psi)$.
 - (ii). Ψ never induces less QuD-ambiguity than Φ , meaning, for no utterance U and insertion site S in U in which Φ or Ψ can be inserted $Q(U, S, \Psi) \prec Q(U, S, \Phi)$.

A covert operator being “preferred by the grammar” does not mean that every grammar must include that operator; a grammar may instead rely exclusively on overt operators, even when semantically similar. Rather, it means that, whenever a grammar includes a covert operator of the relevant kind, that operator will be the optimal one as per (13). Given the present, simplified model of evoked QuDs, (13) should predict $\Phi=pex$ to be preferred over $\Psi=only$, which implies that, if a language possesses a covert exhaustifier, it should be of the *pex* type. Let us sketch a demonstration of this claim. First, (13i) is verified by simple *some* utterances: we already showed that, in sentences such as (1a), $Q(some, root, pex) \prec Q(some, root, only)$. Second, systematically verifying (13ii)—a universal statement over all utterances—would require a more

complete, compositional model of QuDs evoked by complex sentences,¹³ and of their interaction with presupposition projection. Without diving into the details, it seems reasonable to assume that questions evoked by complex sentences are derived from those evoked by their subcomponents. Moreover, increasing structural complexity should refine QuD partitions by subdividing cells rather than fusing them. Under these assumptions, and given that QuDs evoked by a weaker scalemate p and by $pex(p)$ share a cell ($\neg p$), whereas those evoked by p and $only(p)$ do not, more complex utterances involving p will either yield incomparable sets of matching cells, or sets giving an advantage to *pex* over *only*.¹⁴

Before moving on, let us note that our proof of (13i) (a proof of existence) extends to any pair of scalemates $\langle p, p^+ \rangle$, each evoking a polar QuD, and such that $p^+ \models p$. In all such configurations, p 's and $pex(p)$'s QuDs share a $\neg p$ cell, while p 's and $only(p)$'s QuDs share no cell. It follows that $Q(p, root, pex) \prec Q(p, root, only)$, meaning, *pex* creates less QuD-ambiguity when applied to p than *only* does. This generalization also covers “reversed” scales such as $\langle \neg\forall, \neg\exists \rangle$. This is because $pex(\neg\forall)$ typically asserts $\neg\forall$ and presupposes $\neg\neg\exists \equiv \exists$, whereas $only(\neg\forall)$ can be taken to assert $\exists$ and presuppose $\neg\forall$.¹⁵ Moreover, an utterance of $\neg\forall$ can evoke a $\{\forall, \neg\forall\}$ QuD. Together, these observations imply that $pex(\neg\forall)$ evokes a $\{\forall, \exists \wedge \neg\forall\}$ QuD, which shares a $\forall$ cell with the prejacent's QuD; whereas $only(\neg\forall)$ evokes a $\{\exists \wedge \neg\forall, \neg\exists\}$ QuD, which shares no cell with the prejacent's QuD. Since $\emptyset \subset \{\forall\}$, we derive $Q(not\ all, root, pex) \prec Q(not\ all, root, only)$, and the above generalization about $\langle p, p^+ \rangle$ scalemates is indeed verified on the “reversed” $\langle \neg\forall, \neg\exists \rangle$ scale.

This next section explores the predictions of (13) when more complex implicit QuDs are considered, separately, and then, simultaneously.

4. Expanding the pool of potential QuDs

4.1. Three-cell partitions induced by scalemates

An obvious extension of our toy QuD model integrates three-way, Hamblin-style partitions of the CS induced by scalar pairs like $\langle \exists, \forall \rangle$. This is exemplified by the question-answer pair in (14), in which the $\exists \wedge \neg\forall$ cell is made explicitly relevant. In such cases, *all*, *some*, and *only*

¹³Such a model is sketched for a fragment of the language in Hénnot-Mortier (2025).

¹⁴Here is the sketch of a proof, assuming that no presupposition beyond those contributed by the relevant exhaustifiers are at play. Locally, $\exists$, $pex(\exists)$ and $only(\exists)$ evoke the QuDs $\{\neg\exists, \exists\}$, $\{\neg\exists, \exists \wedge \neg\forall\}$, and $\{\exists \wedge \neg\forall, \forall\}$ respectively. It is assumed that, when embedded within the same larger environment E , these expressions evoke potentially finer-grained QuDs derived via the same partition function $\mathcal{P}$. $\mathcal{P}$ maps a cell c (set of worlds) to a set of sub-cells forming a partition of c . Accordingly, $E(\exists)$, $E(pex(\exists))$ and $E(only(\exists))$ evoke $\mathcal{P}(\neg\exists) \cup \mathcal{P}(\exists)$, $\mathcal{P}(\neg\exists) \cup \mathcal{P}(\exists \wedge \neg\forall)$, and $\mathcal{P}(\exists \wedge \neg\forall) \cup \mathcal{P}(\forall)$, respectively. From this, it follows that the set of matching cells between $E(\exists)$'s QuD and $E(pex(\exists))$'s QuD is a superset of $\mathcal{P}(\neg\exists)$, i.e. contains cells overlapping with $\neg\exists$. By contrast, the set of matching cells between $E(\exists)$'s QuD and $E(only(\exists))$'s QuD does not contain any cell overlapping with $\neg\exists$, simply because $\mathcal{P}(\exists \wedge \neg\forall) \cup \mathcal{P}(\forall)$ ($E(only(\exists))$'s QuD) does not overlap with $\neg\exists$. Consequently, the set of matching cells between $E(\exists)$'s QuD and $E(only(\exists))$'s QuD cannot contain the set of matching cells between $E(\exists)$'s QuD and $E(pex(\exists))$'s QuD, meaning, *only* cannot induce less QuD-ambiguity than *pex* in E . This result could in principle be undermined if the CS is known to entail $\exists$, since the advantage of *pex* over *only*, materialized by the set of matching cells $\mathcal{P}(\neg\exists)$, would then disappear. However, such contexts are unlikely to license the use of (*pex*) *some* in the first place.

¹⁵The infelicity of *only not all* is set aside here, since we can focus our attention on covert operators.

some constitute equally felicitous answers.

- (14) Did Jo eat **all**, or just **some** of the cookies?

QuD:

$\forall$	$\exists \wedge \neg\forall$	$\neg\exists$
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Jo ate { all[✓], some[✓], only some[✓] }

Based on (14), both $\exists$ and $\forall$ appear compatible with an overt three-cell QuD of the form $\{\forall, \exists \wedge \neg\forall, \neg\exists\}$. We assume, then, that in such configurations both $\exists$ and $\forall$ evoke this same three-cell QuD. Given how evoked QuDs and presuppositions are thought to interact, we conclude that $pex(\exists)$ may evoke a $\{\exists \wedge \neg\forall, \neg\exists\}$ QuD, obtained via accommodation of $\neg\forall$ onto the three-cell QuD evoked by $\exists$. Similarly, $(c)only(\exists)$ may evoke a $\{\forall, \exists \wedge \neg\forall\}$ QuD, obtained via accommodation of $\exists$ onto the three-cell QuD evoked by $\forall$. These QuD structures happen to match those derived for these operators in Section 2 under the assumption that $\exists$ and $\forall$ both evoked “polar” (though distinct) questions.

The QuDs evoked by $pex(\exists)$ and $(c)only(\exists)$ then each share the same cell $(\exists \wedge \neg\forall)$ with the QuD evoked by $\exists$. Thus, their respective sets of matching cells with the QuD evoked by their preadjacent are equal: $M(Q(some, root, pex)) = M(Q(some, root, only)) = \{\exists \wedge \neg\forall\}$. Since these sets are equal, neither stands in a *strict* subset relation to the other. It then follows, by (9) and (12), that pex ’s and $only$ ’s potentials for QuD-ambiguity are incomparable in that particular setting—neither $Q(some, root, pex) \prec Q(some, root, only)$ nor $Q(some, root, only) \prec Q(some, root, pex)$ holds.

This shows that three-cell QuD structures of the type in (14) are neutral with respect to the choice between pex and $only$: they do not favor either operator under the QuD-ambiguity metric, and in particular do not support a preference for $only$. The next section considers similar configurations arising when a third alternative is available beyond $\exists$ and $\forall$.

4.2. Cases in which more than two scalemates are relevant

While $\forall$ is arguably the most salient alternative to $\exists$, other alternatives of intermediate strength may also be relevant; for instance *most* (abbreviated *M*). This observation extends to other scales, including $\langle might, may, must \rangle$, $\langle sometimes, often, always \rangle$. How does the existence of intermediate alternatives influence evoked QuDs, and in turn, our model of QuD-ambiguity and its predictions? (15) illustrates question-answer pairs in which *M* is directly relevant to the QuD, producing partitions with more than two cells.

- (15) a. Did Jo eat **most**, or just **some** of the cookies?

QuD:

<i>M</i>	$\exists \wedge \neg M$	$\neg\exists$
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Jo ate { all[?], most[✓], some[✓], only some[✓] }

- b. Did Jo eat **all**, **most**, or just **some** of the cookies?

QuD:

$\forall$	$M \wedge \neg\forall$	$\exists \wedge \neg M$	$\neg\exists$
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Jo ate { all[✓], most[✓], some[✓], only some[✓] }

We will first focus on the competition between pex and $only$ given a $\exists$ -preadjacent, considering

the QuDs in (15a) and (15b) successively.¹⁶ We will then turn to the competition between these operators applied to a *M*-prejacent, given the QuD in (15b). This will ultimately lead to a generalization about the QuD-ambiguity induced by *pex* vs. *only*, whenever Hamblin-style QuDs induced by an arbitrary number of scalemates are considered.

In (15a), $\exists$ and *M*, but not $\forall$, are relevant to the QuD. It is thus reasonable to assume that exhaustifying $\exists$ in that context would lead to infer $\neg M$, instead of the usual and logically weaker $\neg\forall$. This inference is at-issue if produced by *(c)only* and presupposed (yet easily accommodated) if produced by *pex*. Moreover, *some*, *only some*, and *most* in (15a) appear compatible with the contextually provided QuD, $\{M, \exists \wedge \neg M, \neg\exists\}$. So, in that setting, we take $\exists$ and *M* to evoke the same three-cell $\{M, \exists \wedge \neg M, \neg\exists\}$ QuD. Given how evoked QuDs and presuppositions are thought to interact, we conclude that *pex*($\exists$) may evoke a $\{\exists \wedge \neg M, \neg\exists\}$ QuD (obtained by accommodating $\neg M$ on $\exists$'s QuD), while *(c)only*($\exists$) may evoke a $\{M, \exists \wedge \neg M\}$ QuD (obtained by accommodating $\exists$ on *M*'s QuD). This is summarized in Table 4. The QuD evoked by *pex*($\exists$) then shares a $\exists \wedge \neg M$ and a $\neg\exists$ cell with the QuD evoked by $\exists$; while the QuD evoked by *(c)only*($\exists$) shares a *M* and a $\exists \wedge \neg M$ cell with it. Therefore, the sets of matching cells at stake merely overlap, which implies that *pex*'s and *only*'s potentials for QuD-ambiguity are incomparable in this configuration.

Turning to (15b), whereby $\exists$, *M* and $\forall$ are all relevant to the QuD. Again here, it is reasonable to assume that exhaustifying $\exists$ would lead to infer $\neg M$. And in that case again, utterances of *some*, *only some*, and *most* appear compatible with the contextually provided four-cell QuD, $\{\forall, M \wedge \neg\forall, \exists \wedge \neg M, \neg\exists\}$. This allows us to conclude that *pex*($\exists$) and *(c)only*($\exists$) may evoke, respectively, a $\{\exists \wedge \neg M, \neg\exists\}$ and a $\{\forall, M \wedge \neg\forall, \exists \wedge \neg M\}$ QuD after their respective presuppositions get accommodated. This is summarized in Table 5. The QuD evoked by *pex*($\exists$) then shares a $\exists \wedge \neg M$ and a $\neg\exists$ cell with the QuD evoked by $\exists$; while the QuD evoked by *(c)only*($\exists$) shares a $\forall$, a $M \wedge \neg\forall$, and a $\exists \wedge \neg M$ cell with it. Once again, the sets of matching cells at stake merely overlap, which implies that *pex*'s and *only*'s potentials for QuD-ambiguity are, again, incomparable in this configuration.

Table 4: QuD s evoked by $\exists$, *pex*($\exists$) and *only*($\exists$), assuming $\exists$ and *M* are relevant while $\forall$ is not, as in (15a).

Parse	Evoked QuD		
$\exists$	<i>M</i>	$\exists \wedge \neg M$	$\neg\exists$
<i>pex</i> ($\exists$)		$\exists \wedge \neg M$	$\neg\exists$
<i>only</i> ($\exists$)	<i>M</i>	$\exists \wedge \neg M$	

Table 5: QuD s evoked by $\exists$, *pex*($\exists$) and *only*($\exists$), assuming $\exists$, *M* and $\forall$ are relevant, as in (15b).

Parse	Evoked QuD			
$\exists$	$\forall$	$M \wedge \neg\forall$	$\exists \wedge \neg M$	$\neg\exists$
<i>pex</i> ($\exists$)			$\exists \wedge \neg M$	$\neg\exists$
<i>only</i> ($\exists$)	$\forall$	$M \wedge \neg\forall$	$\exists \wedge \neg M$	

Before moving on, let us discuss one final case in which *pex* and *(c)only* apply to *M*. (15b) constitutes a context in which this can be done non-trivially: because $\forall$ is relevant to the QuD, exhaustifying *M* likely produces a $\neg\forall$ inference, made at-issue by *(c)only* and presupposed by *pex*. So *pex*(*M*) can be taken to assert *M* and presuppose $\neg\forall$, and *(c)only*(*M*), to assert $\neg\forall$ and to presuppose *M*. Additionally in (15b), *M* and $\forall$ both appear compatible with the contextually provided four-cell QuD, $\{\forall, M \wedge \neg\forall, \exists \wedge \neg M, \neg\exists\}$. This allows us to derive QuDs for *pex*(*M*)

¹⁶Interestingly, answering with *only most* or *just most* is quite degraded in (15), even when the QuD explicitly targets *M* and $\forall$. We do not address this puzzle here.

and $(c)only(M)$, by accommodating, respectively, $\neg\forall$ and M on the $\{\forall, M \wedge \neg\forall, \exists \wedge \neg M, \neg\exists\}$ QuD. The outcome of this process is summarized in Table 6, which shows that the QuD evoked by $pex(M)$ shares a $M \wedge \neg\forall$, a $\exists \wedge \neg M$, and a $\neg\exists$ cell with the QuD evoked by M ; while the QuD evoked by $(c)only(M)$ shares a $\forall$, and a $M \wedge \neg\forall$ cell with it. This pattern appears similar to that of Table 5, the only difference being that it is slightly shifted to the left (which stems from the fact that the prejaçant now at stake, M , is slightly stronger than $\exists$). Once again, the mere overlap between the sets of matching cells at stake allows us to conclude that pex 's and $only$'s potentials for QuD-ambiguity are incomparable in that setting.

Table 6: QuD s evoked by M , $pex(M)$ and $only(M)$, assuming $\exists$, M and $\forall$ are relevant, as in (15b).

Parse	Evoked QuD			
M	$\forall$	$M \wedge \neg\forall$	$\exists \wedge \neg M$	$\neg\exists$
$pex(M)$		$M \wedge \neg\forall$	$\exists \wedge \neg M$	$\neg\exists$
$only(M)$	$\forall$	$M \wedge \neg\forall$		

The patterns exhibited by Tables 4, 5 and 6 appear generalizable to an arbitrary number of scalarly ordered alternatives inducing a Hamblin-style QuD; for any non-endpoint alternative p on the scale, $pex(p)$ and $(c)only(p)$ evoke QuDs that overlap only on the cell corresponding to the exhausted meaning of p according to that scale, conflating presupposition and assertion.¹⁷ At first glance, this result may seem negative, just like the one established in the previous section: it implies that increasing QuD complexity systematically leaves pex and $only$ incomparable. However, it is in fact consistent with, and supportive of, the claim that pex minimizes QuD-ambiguity overall. Having already shown that pex can outperform $only$ for *some* utterance/QuD pairing (condition (13i)), it remains only to verify that $only$ is *never* preferred across alternative utterance/QuD pairings (condition (13ii)). The results of this section and the previous one jointly support this conclusion.

This line of reasoning, however, has a shortcoming: it holds on a case-by-case basis. In each case, a bare utterance ($\exists$, M , or $\forall$) was assumed to evoke a single fixed QuD, which, when intersected with the relevant presuppositions, would yield one QuD for the pex -ed counterpart of the utterance, and one QuD for the $(c)only$ -ed counterpart of the same utterance. The next section examines whether the current proposal is still valid under the assumption that a given utterance may evoke multiple candidate QuDs *simultaneously*—in the form of a *set* of QuDs such as $\{\{\exists, \neg\exists\}, \{\forall, \exists \wedge \neg\forall, \neg\exists\}, \{M, \exists \wedge \neg M, \neg\exists\}, \{\forall, M \wedge \neg\forall, \exists \wedge \neg M, \neg\exists\}\}$.

4.3. Generalizing to multiple implicit QuDs

Instead of reasoning in terms of a single fixed QuD for each parse, this section explores the possibility that a given utterance may simultaneously evoke multiple candidate QuDs. For simplicity, we assume that the M alternative is not relevant. An utterance of $\exists$ may then evoke either a “polar” QuD of the form $\{\exists, \neg\exists\}$ (discussed in isolation in Sections 2 and 3), or a three-cell

¹⁷I.e. $p \wedge \neg p^+$, with p^+ the immediately stronger alternative to p on the relevant scale.

Covert operators are picked to minimize QuD-Ambiguity: The view from *pex* and *only*

Hamblin-style QuD induced by $\exists$ along with its scalemate $\forall$, of the form $\{\forall, \exists \wedge \neg\forall, \neg\exists\}$ (discussed in isolation in Section 4.1). Likewise, an utterance of $\forall$ may evoke either a polar $\{\forall, \neg\forall\}$ QuD, or a three-cell $\{\forall, \exists \wedge \neg\forall, \neg\exists\}$ QuD. Let us refer to these four different partitions of the CS (two for $\exists$, two for $\forall$) as, respectively, $Q_{\exists,1}$, $Q_{\exists,2}$, $Q_{\forall,1}$, $Q_{\forall,2}$. Note that $Q_{\exists,2} = Q_{\forall,2}$. QuDs for *pex*($\exists$) and *conly*($\exists$) can then be derived from these candidate QuDs. Recall that a QuD for *pex*($\exists$) is derived by intersecting a QuD for $\exists$ (*pex*($\exists$)’s assertion) with $\neg\forall$ (*pex*($\exists$)’s presupposition); this is done on $Q_{\exists,1}$ and $Q_{\exists,2}$ in Figure 4. The outcome is independent of the candidate QuD selected for $\exists$: in every case, the QuD derived for *pex*($\exists$) is $\{\exists \wedge \neg\forall, \neg\exists\}$.

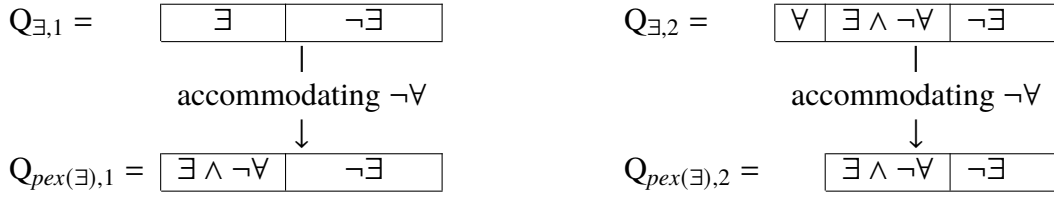


Figure 4: Accommodating $\neg\forall$ (*pex*($\exists$)’s presupposition) on candidate QuDs evoked by $\exists$.

Similarly, a QuD for *conly*($\exists$) is derived by intersecting a QuD for $\forall$ (*conly*($\exists$)’s assertion) with $\exists$ (*conly*($\exists$)’s presupposition); this is done on $Q_{\forall,1}$ and $Q_{\forall,2}$ in Figure 5. As in the case of *pex*, the outcome is independent of the candidate QuD selected for $\exists$: in every case, the QuD derived for *conly*($\exists$) is $\{\forall, \exists \wedge \neg\forall\}$.



Figure 5: Accommodating $\exists$ (*conly*($\exists$)’s presupposition) on candidate QuDs evoked by $\forall$.

Let us now evaluate the QuD-ambiguity associated with *pex* and *conly*, given their respective evoked QuDs and the multiple candidate QuDs evoked by the bare prejaent $\exists$. Recall that, according to (11), two operators can be compared through the lens of to QuD-ambiguity by examining the sets of matching cells between the QuDs they evoke and those evoked by the prejaent. To compute sets of matching cells, (9-11) proposed collecting all the QuDs evoked by parses of the target utterance with and without the operator, and taking their joint intersection. Consider first *pex*. The set of QuDs evoked by either $\exists$ or *pex*($\exists$) is now $\{Q_{\exists,1}, Q_{\exists,2}, Q_{pex(\exists),1}\}$ (since $Q_{pex(\exists),1}=Q_{pex(\exists),2}$). The resulting set of matching cells is then $Q_{\exists,1} \cap Q_{\exists,2} \cap Q_{pex(\exists),1} = \{\neg\exists\}$, meaning, $\neg\exists$ is the only cell shared among the three QuDs. Turning to *conly*, the set of questions evoked by either $\exists$ or *conly*($\exists$) is now $\{Q_{\exists,1}, Q_{\exists,2}, Q_{conly(\exists),1}\}$ (since $Q_{conly(\exists),1}=Q_{conly(\exists),2}$). The resulting set of matching cells is then $Q_{\exists,1} \cap Q_{\exists,2} \cap Q_{conly(\exists),1} = \emptyset$. Indeed, $Q_{\exists,1}$ and $Q_{conly(\exists),1}$ share no cell, and therefore neither do all three QuDs taken together.

Since $\emptyset \subset \{\neg\exists\}$, we derive $Q(\text{some}, \text{root}, \text{pex}) \prec Q(\text{some}, \text{root}, \text{only})$, just as we did in Section 3. Thus, allowing parses to evoke multiple implicit QuDs simultaneously does not appear to undermine the predictions of (12), and the fact that *pex* should be overall favored as covert operator.

This result can be shown to extend to cases in which even more Hamblin-style, alternative-induced QuDs are considered simultaneously. As long as every candidate QuD evoked by $\exists$ involves a $\neg\exists$ cell (a plausible assumption), any QuD derived for $\text{pex}(\exists)$ will preserve (and thus “match”) that cell, whereas QuD(s) derived for $\text{only}(\exists)$ will not.¹⁸ Consequently, *pex*’s and *only*’s potentials for QuD-ambiguity will be either incomparable, or result in a preference for *pex*. This conclusion is noteworthy, because it relies only on minimal assumptions about the QuDs evoked by $\exists$. No parallel assumptions are required about the QuDs evoked by $\forall$ for instance. Even more generally, the same reasoning extends to richer scales of the form $\langle \dots, p, p^+, \dots \rangle$.

5. Discussion and outlook

This paper developed and formalized the idea that semantically similar objects, such as *pex* and *only*, somehow “compete” for covertness, in the sense that languages follow systematic principles that permit at most one of them to be covert. It was further argued that, whenever a covert operator is grammaticized, the grammar favors the one that minimizes the QuD-ambiguity induced by its covertness. The following four paragraphs discuss the broader intuitions behind our characterization of QuD-ambiguity, the conceptual benefits and theoretical implications of this definition, the limitations encountered in deriving its predictions (and whether these limitations are truly problematic), and, lastly, broader typological consequences of the proposal.

QuD-ambiguity was defined as a relative and partial measure, based on sets of matching cells between potential implicit questions. These sets are obtained by identifying the cells shared across all candidate QuDs evoked either by a prejacent or by that prejacent enriched with the relevant covert operator. In set-theoretic terms, this amounts to *intersecting* the questions belonging to the *union* of two sets: (i) the set of questions evoked by the prejacent, and (ii) the set of questions evoked by the operator-enriched prejacent. From a game-theoretic perspective, computing and comparing sets of matching cells to assess QuD-ambiguity can thus be understood as implementing a min-max strategy: one attempts to extract the maximal amount of relevant information (cells) that candidate QuDs have in common, making minimal assumptions about the origin and derivation of these QuDs (i.e. actively considering the possibility of QuD optionality, and parsing ambiguities). On a more operational level, gathering all the QuDs evoked by the two parses (with and without the target covert operator) in the same set prior to computing their common denominators (matching cells) was probably the most straightforward implementation of the concept of “set of matching cells”. Whether it is the most *empirically* accurate implementation however (especially under QuD optionality), is left for future work.¹⁹

¹⁸They will in fact exclude the $\neg\exists$ subspace of the CS altogether, due to *only*($\exists$)’s presupposition ($\exists$).

¹⁹Indeed, assuming that any utterance may evoke multiple QuDs, merging all the QuDs evoked by a prejacent *p* with those evoked by, say, *pex*(*p*), erases the derivational links that preexisted between a QuD for *p* and the QuD for *pex*(*p*) that was derived from it by accommodation/intersection. In that sense, our pooling of *all* the candidate QuDs in the same set before finding the matching cells between them erases potentially relevant information, taking the form of specific QuD-pairings.

The covertness asymmetry between *pex* and *only* was then cashed out by comparing how the insertion of these operators in any given utterance “shifts” the QuD originally evoked by that utterance. *pex* was shown to be less of a QuD “shifter” than *(c)only*, since its insertion tends to preserve more cells from the prejacent’s evoked QuD. This contrast between the two operators can be traced back to how *pex* and *(c)only* divide up the work between presupposition and assertion: *pex* keeps the assertion of its prejacent at-issue, and contributes additional SIs presuppositionally, while *only* systematically backgrounds its prejacent’s original assertion, contributing additional SI-like inferences at the assertoric level. Connecting this merely *descriptive* difference between *pex*’s and *only*’s lexical entries to implicit QuDs (and the concept of covertness-induced QuD-ambiguity), made way for a more explanatory account of the observed pattern. It is also worth mentioning that adopting *exh* instead of *pex* would complicate the analysis developed in this paper. Unlike *pex*, *exh* is typically assumed to assert, not only its prejacent, but rather, the conjunction of its prejacent with SIs. As a result, the assertion of *exh(p)* is not aligned with its prejacent in the same way as in the *pex* case. It is therefore less clear how the QuDs evoked by *exh(p)* could be “closer” to those evoked by *p* than the QuDs evoked by *only(p)* are. This difficulty constitutes an indirect argument in favor of *pex* over *exh*: within the present framework, treating the attested covert exhaustifier as a semantic mirror image of *only* makes its covert status less puzzling and more naturally integrated into the QuD-ambiguity minimization story.

This paper focused on showing inherent differences between *pex* and *(c)only* at the matrix level, demonstrating that, regardless of which exact set of implicit QuDs or which scalar structure was considered, *pex*’s covertness would either induce less QuD-ambiguity than *only*’s, or the degrees of QuD-ambiguity induced by these operators would be incomparable. However, a full generalization of our demonstration to complex embedded environments (potentially involving intricate presupposition projection patterns and multiple insertion sites for exhaustifiers like *pex* and *(c)only*) was only impressionistically sketched. Although this constitutes a fundamental gap in the argumentation, one can wonder whether, on more empirical grounds, grammars are expected to optimize QuD-ambiguity uniformly across *all* utterances (including the rarest and most complex ones), or whether simple matrix utterances (that we covered extensively here) constitute the core benchmark for such grammars, at least when it comes to selecting preferred covert operators.

Let us close this paper with our account’s main typological prediction, which is that, if SIs—whether Neo-Gricean, or grammatical, materialized by *pex*—are present in every single language, no language should display a covert variant of *only* (*conly*). Moreover, if SIs are grammatical, some languages may display *pex*, some others, *opex*, and some languages may in fact display both simultaneously. The configuration that our approach definitely excludes, is an *opex/conly* configuration. The exact inventory of exhaustifiers displayed by a language, given these constraints, is a function of the language’s tolerance for ambiguity, and sensitivity to prolixity. This raises a number of questions: why do so many languages display *pex* instead of *opex* (along with *only*)? Is this dominant pattern solely due to cost-related considerations outweighing a need for disambiguation? Are there any other potentially overlooked advantages tied to covert (and hence optional) operators? Relatedly, what treatment should be assigned to more complex (and hence costly!) constructions sharing similarities with *pex*, e.g. clefts in English? Future typological work should determine what the actual quantitative trends are in the domain of (c)overt exhaustifiers, and whether these trends can inform our model and predictions in any way. Lastly, if this

approach is on the right track, it should be expected to apply to other sets of (c)overt operators characterized by a similar-enough semantics. A domain to explore may for instance be that of (bare) plurals.

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